Ali Aan 23I-0708

https://github.com/aliaankhowaja/Cricket

1. Game Logic and Working Mechanism

The game is a single-player 2D cricket batting game. The player bats for a fixed match duration of 2 overs (12 balls) with a maximum of 2 wickets. On each ball, the player selects either Aggressive or Defensive batting style. A slider continuously moves across a probability-based power bar. When the player clicks anywhere on the game container, the slider stops and its position determines the outcome of the ball.

Each batting attempt increments the ball count. The game ends when all 12 balls are bowled or both wickets are lost. A scoreboard displays total runs, wickets fallen, and balls/overs remaining, updated after every ball. A restart button resets all game state.

2. Mapping of Probabilities to the Power Bar

Each batting style defines a probability distribution across seven outcomes: Wicket, 0, 1, 2, 3, 4, and 6 runs. The power bar is divided into colored segments, each occupying a width proportional to its assigned probability. The slider position (0.0 to 1.0) is compared against cumulative probability boundaries to determine the outcome.

Defensive Style

Outcome	Probability	Cumulative Range
Wicket	0.10	0.00 – 0.10
0 Runs	0.30	0.10 – 0.40
1 Run	0.40	0.40 – 0.80
2 Runs	0.06	0.80 – 0.86
3 Runs	0.05	0.86 – 0.91
4 Runs	0.05	0.91 – 0.96
6 Runs	0.04	0.96 – 1.00
Total	1.00	

Aggressive Style

Outcome	Probability	Cumulative Range
Wicket	0.40	0.00 – 0.40
0 Runs	0.05	0.40 – 0.45
1 Run	0.05	0.45 – 0.50
2 Runs	0.05	0.50 – 0.55
3 Runs	0.10	0.55 – 0.65
4 Runs	0.20	0.65 – 0.85
6 Runs	0.15	0.85 – 1.00
Total	1.00	

3. Implementation of Animations

All animations are implemented using CSS keyframes and JavaScript class toggling.

Before each shot, the ball element travels from the bowler's end toward the batsman using the bowl-and-hit or bowl-to-stumps keyframe animation, simulating a realistic delivery arc.

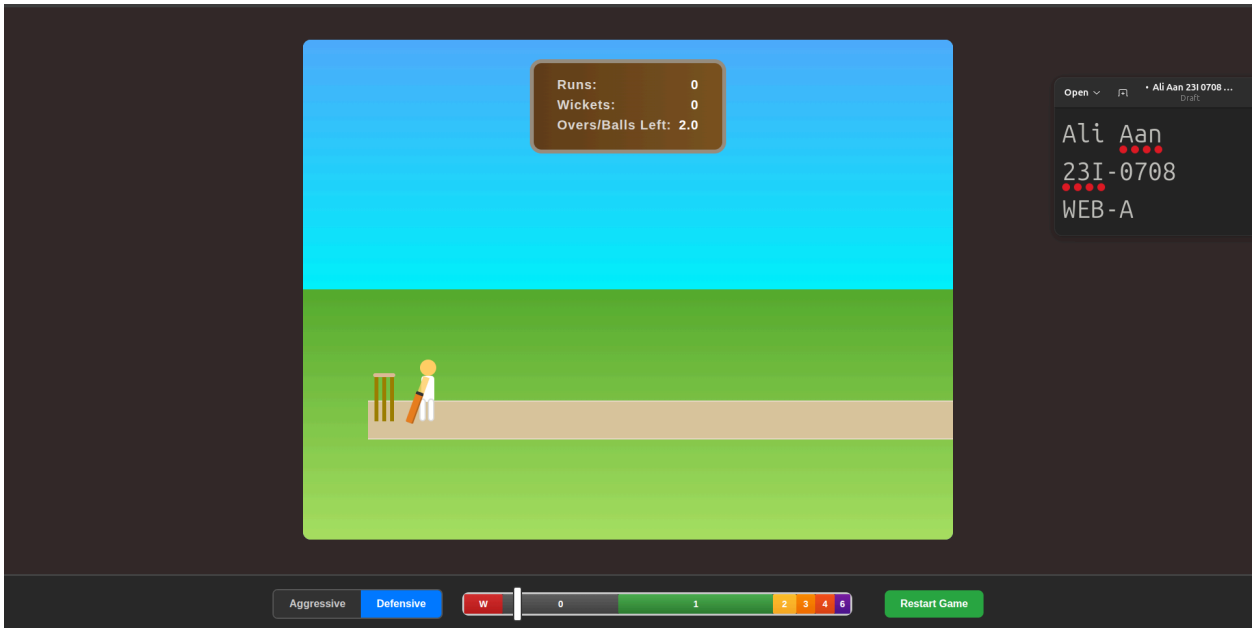
When the player scores runs, the ball moves toward the batsman and continues off-screen after the hit. The batsman's arm swings via the swing-bat keyframe, timed with a delay to synchronize with the ball's arrival.

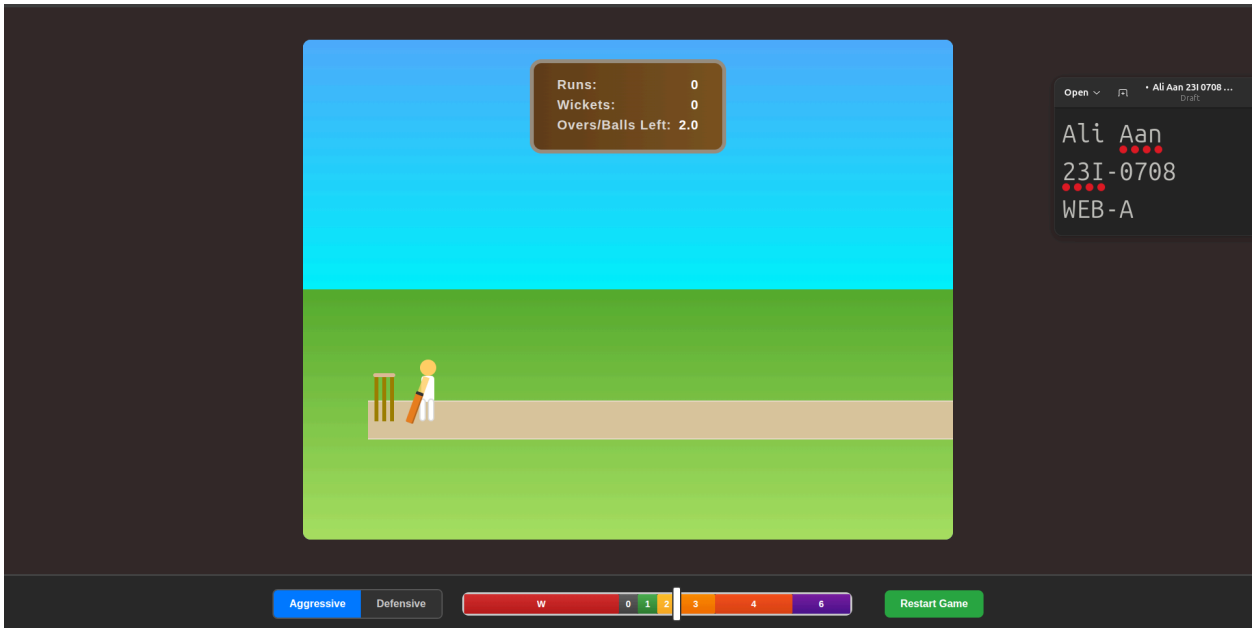
When a wicket falls, the ball travels directly into the stumps. The stumps rotate and translate to simulate being knocked over, and the bails fly off with a fade-out effect.

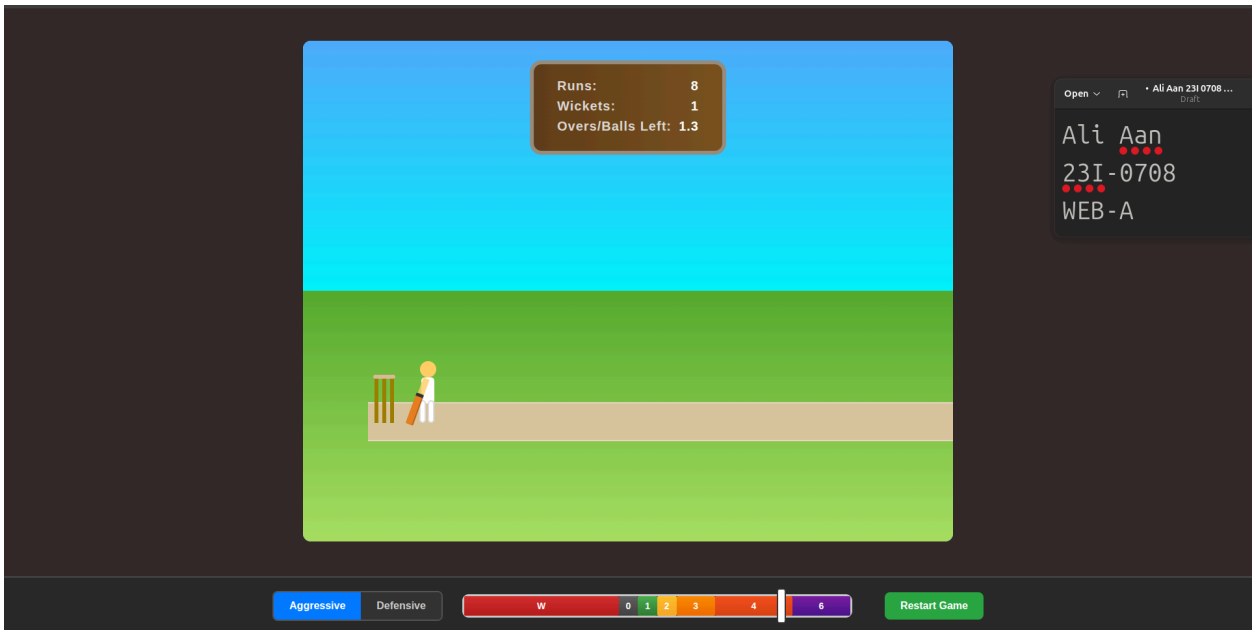
After each ball, the run result (e.g., "4", "OUT!") scales up and fades in at the center of the game screen using a CSS burst animation.

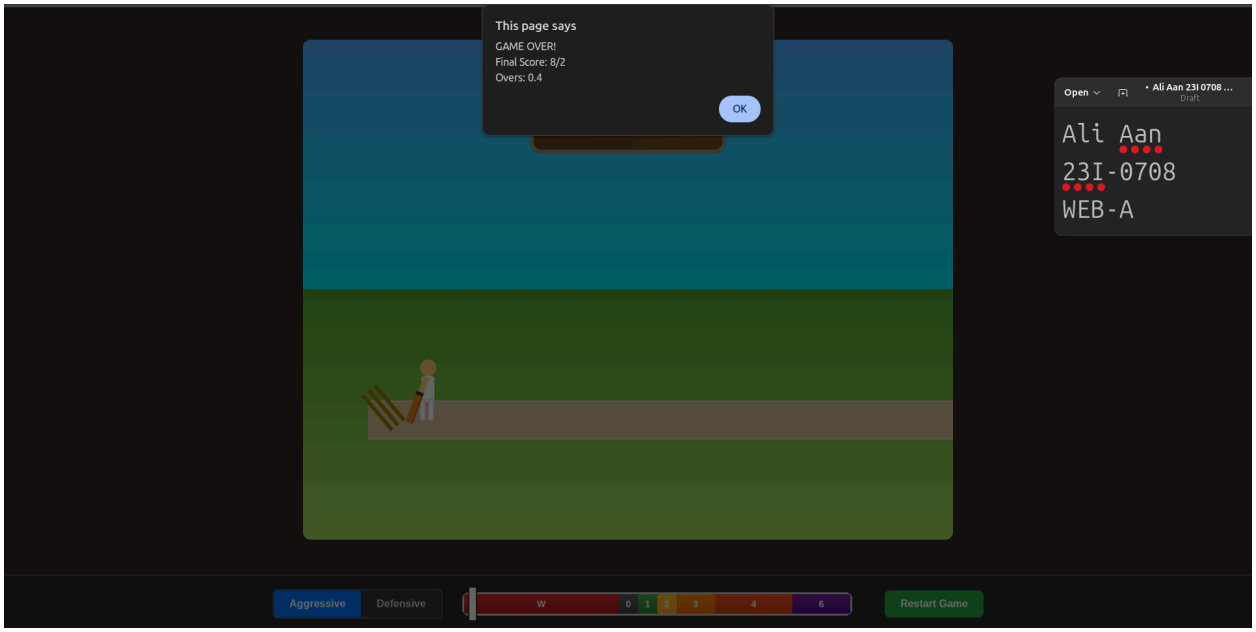
All animations are reset after each ball using classList.remove() and a forced reflow (offsetWidth access) to allow re-triggering on subsequent deliveries.

4. Screenshots


Runs: 0
Wickets: 0
Overs/Balls Left: 2.0
Open
• Ali Aan 23I 0708 ...
Draft
Ali Aan
23I-0708
WEB-A
Aggressive
Defensive
W
0
1
2
3
4
6
Restart Game


Runs: 0
Wickets: 0
Overs/Balls Left: 2.0
Open
• Ali Aan 23I 0708 ...
Draft
Ali Aan
23I-0708
WEB-A
Aggressive
Defensive
W
0
1
2
3
4
6
Restart Game


Runs: 8
Wickets: 1
Overs/Balls Left: 1.3
Open
• Ali Aan 23I 0708 ...
Draft
Ali Aan
23I-0708
WEB-A
Aggressive
Defensive
W
0
1
2
3
4
6
Restart Game


This page says
GAME OVER!
Final Score: 8/2
Overs: 0.4
OK
Open
Ali Aan 23I 0708 ...
Draft
Ali Aan
23I-0708
WEB-A
Aggressive
Defensive
W
0
1
2
3
4
6
Restart Game